

Time Series Analysis - Singapore Total Fertility Rate and Total Live Births

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Abstract

Over the course of the last century, Singapore's demographic transition remains as one of the most rapid ever recorded, with the Total Fertility Rate (TFR) declining from 5.76 in 1960 to less than 1.0 by 2024. This paper features a complete time series analysis of two interrelated population metrics: TFR and Total Live Births (TLB). These are scraped from the Singapore Department of Statistics' annual data, ranging from 1968 to 2024. After an established Data Analysis Lifecycle, paired with the Box-Jenkins identification cycle, a log-differencing transformation is required to achieve stationarity. This is also furthermore confirmed by ADF and KPSS tests. Autocorrelation (ACF) and partial autocorrelation (ACF) functions applied to our stationary series prompts for an $ARIMA(p, 1, q)$ candidate model(s) fitted with a logarithmic Box-Cox transform. Model selection verification using AICc identifies the specification with best fit for each series; residual diagnostics aim to replicate white noise behaviour. Out-of-sample forecasts over a subsample of 2013–2024 are evaluated against observed values, with performance assessed via RMSE, MAE, MAPE, and Theil's U. The results post-evaluation confirm the particular difficulty of accurately forecasting sub-replacement fertility prediction mapping, likely due to structural policy interventions within Singapore and/or the impact of COVID-19. This motivates additional extension processes including ARIMA-GARCH and state-space modelling.

Keywords: Total Fertility Rate; Total Live Births; ARIMA; Box-Jenkins; Stationarity; Singapore

1 Introduction

- Demographic transition
- Discuss variables
- Discuss significant change in variables, and why. EDA
- Introduce research questions: ‘which policy interventions over 1960–2024 justify the significant structural deviations in TFR and TLB?’, ‘can ARIMA models trained on the 1960–2012 window project the 2013–2024 period?’, ‘what (and how do) demographic mechanisms account for the recent significant drop in TFR?’
- State IED Box-Jenkins cycle as diagnostic tool

2 Data

2.1 Data Sourcing and Structuring

The data used throughout this report are sourced from *SingaporeCensus.csv*, a population census published by the Singapore Department of Statistics. The raw comma-separated-value file is stored in wide format, i.e. the years 1960–2024 are column headers. Two rows are extracted – one for TFR and one for TLB – and reshaped to long format by using `pivot_longer()`. The resulting cleaned dataset contains 65 annual observations per series with no missing values.

For forecasting evaluation, the full series is partitioned into a training windows of $n = 53$ observations (from 1960 to 2012) and a complementary windows of $h = 12$ observations (2013 to 2024).

2.2 Statistical Summary

Table 1: Summary statistics for TFR and TLB (1960–2024).

Statistic	TFR	TLB
Min	0.97	33541.00
Q1	1.26	39570.00
Median	1.62	42484.00
Mean	2.06	44359.63
Q3	2.07	48577.00
Max	5.76	61775.00
NAs	0.00	0.00

TFR ranges from an initial post-war high of 5.76, to a final minimum of 0.97, which is much lower than the ‘replacement threshold’ of 2.1 (Espenshade, Guzman, and Westoff 2003). The considerable difference between the mean of 2.06 and median of 1.62 is reflective of a strongly right-skewed distribution produced by the pre-transitional years.

2.3 Time Series Plots

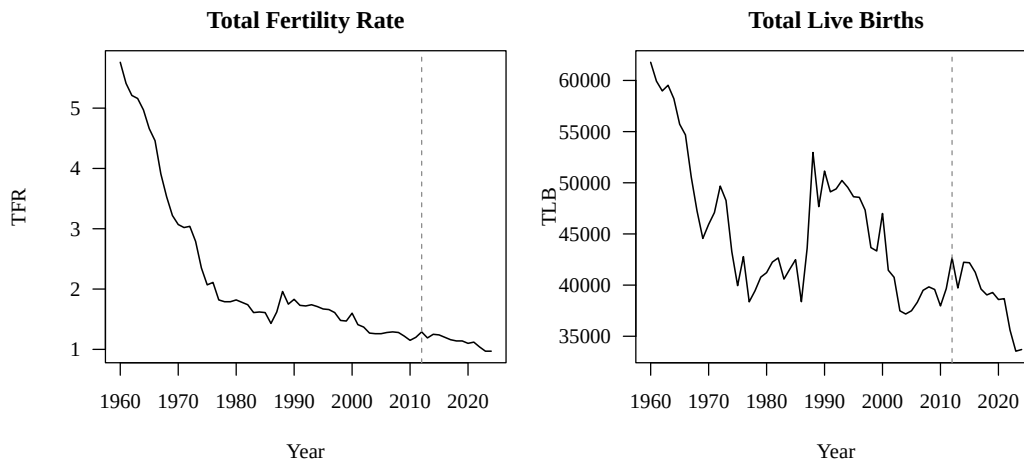


Figure 1: TFR (left) and TLB (right), from 1960–2024. The dashed vertical line indicates the train-test data partitioning.

Figure 1 indicates two salient features shared by both series. For one, a strong uniform downward trend persists over the full sample, which immediately rules out non-transformative modelling approaches. For two, the variance of the fluctuations around the trend is considerably higher during earlier years (1960–1980) instead of during the sub-replacement period post-1985. This is indicative of a multiplicative error structure rather than an additive one, which prompts for a logarithmic variance stabilisation method.

3 Context and Methodology

3.1 Transformation and Stationarity

A time series $\{X_t\}$ is said to be *weakly stationary* if its unconditional mean $\mu = E[X_t]$ is constant and its autocovariance function $\gamma(h) = \text{Cov}(X_t, X_{t-h})$ depends only on lag h , rather than time t (Box and Jenkins 1970). Both raw series are in violation of this condition; shown is a persistent downward trend (i.e. a non-constant mean) and heteroskedastic fluctuations (i.e. non-constant variance).

Since variance appears proportional to level, a logarithmic transformation $W_t = \log X_t$ is applied to stabilise variance. The log series retains a trend, prompting for a single first-difference to produce a stationary working series:

$$Y_t = (1 - B) \log X_t = \log X_t - \log X_{t-1} = \log \left(\frac{X_t}{X_{t-1}} \right),$$

where B is the backward shift operator ($BX_t = X_{t-1}$). In this context, Y_t is interpreted as the log-annual growth rate of the series. Stationarity of Y_t is assessed using two statistical tests:

- **Augmented Dickey-Fuller (ADF)** (Dickey and Fuller 1979): tests H_0 (unit root) against H_1 (stationary). A small p -value supports rejection of the unit-root null.
- **KPSS** (Kwiatkowski et al. 1992): tests H_0 : (trend-)stationary against H_1 (unit root). A large p -value (non-rejection) supports the stationarity hypothesis.

Using both tests simultaneously ensures less ambiguity that individual tests can experience; due to low power or size distortion, for instance.

3.2 ARIMA Model specification

An ARIMA(p, d, q) model applied to the log-transformed series $\log X_t$ (so that $d = 1$ acts on the log series) is:

$$\phi(B)(1 - B) \log X_t = c + \theta(B)\varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2),$$

where $\text{WN}(0, \sigma^2)$ denotes a white-noise sequence (an uncorrelated process with zero mean and constant variance σ^2). The AR and MA polynomials are:

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p, \quad \theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q.$$

This model is *causal* (stationary) if all roots of $\phi(B) = 0$ lie strictly outside the unit circle; additionally, it is *invertible* if all roots of $\theta(B) = 0$ lie strictly outside the unit circle.

To specify this, the `forecast` R package features `Arima(x, order = c(p, 1, q), lambda = 0)`, where `lambda = 0` applies the logarithmic Box-Cox transform internally, and automatically back-transforms forecasts to the original scale.

3.2.1 Model identification

Given a stationary series Y_t , the **autocorrelation function** (ACF) at lag k is:

$$\rho_k = \frac{\text{Cov}(Y_t, Y_{t-k})}{\text{Var}(Y_t)},$$

and the **partial autocorrelation function** (PACF) α_k measures the linear dependence between Y_t and Y_{t-k} after eliminating the effect of $Y_{t-1}, \dots, Y_{t-k+1}$. The Box-Jenkins identification table is:

Model	Theoretical ACF	Theoretical PACF
AR(p)	Geometric decay/oscillation	Cuts off after lag p
MA(q)	Decays after lag q	Geometric decay/oscillation
ARMA(p, q)	Decays after lag $q - p$	Decays after lag $p - q$

Approximate 95% bounds on the sample ACF/PACF are $\pm 1.96/\sqrt{n}$, where n is the length of the differenced series.

3.2.2 Model selection

Candidate models are compared using the corrected Akaike Information Criterion:

$$\text{AICc} = -2\hat{\ell} + 2k + \frac{2k(k+1)}{n-k-1},$$

where $\hat{\ell}$ is the maximised log-likelihood and k is the number of free parameters. AICc is typically preferred over AIC for small n (Burnham and Anderson 2002) as a stronger model complexity penalty reduces the risk of overfit and over-parameterisation.

3.3 Residual Diagnostics

An accurate, well-specified model should yield residuals $\hat{\varepsilon}_t$ that resemble white noise. This is assessed using three complementary tools:

1. **ACF of residuals.** All sample autocorrelations should lie within the 95% (i.e., $\pm 1.96/\sqrt{n}$);
2. **Ljung-Box test** (Ljung and Box 1978). The portmanteau statistic

$$Q^* = n(n+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{n-k}$$

is compared to $\chi^2(m-p-q)$ under H_0 (i.e., no residual correlation prior to lag m). The degrees of freedom correction `fitdf` = $p + q$ accounts for parameters that are influenced by estimation error. Omitting it (setting `fitdf` = 0) artificially ‘inflates’ the χ^2 test statistic;

3. **Residual histogram:** valid Gaussian prediction intervals require residuals that are approximately Normal.

3.4 Forecast Accuracy Measurement

Point-forecast performance over our ‘hold-out’ period (2013–2024) is measured by:

$$\text{RMSE} = \sqrt{\frac{1}{h} \sum_{t=1}^h e_t^2}, \quad \text{MAE} = \frac{1}{h} \sum_{t=1}^h |e_t|, \quad \text{MAPE} = \frac{100}{h} \left| \frac{e_t}{X_t} \right|$$

where $e_t = \hat{X}_t - X_t$. As mentioned, **Theil’s U** compares model RMSE to that of the naive (that is, the random walk) forecast; $U < 1$ indicates the model outperforms the initial naive benchmark.

4 Exploratory Analysis

4.1 Log Transformation Assessment

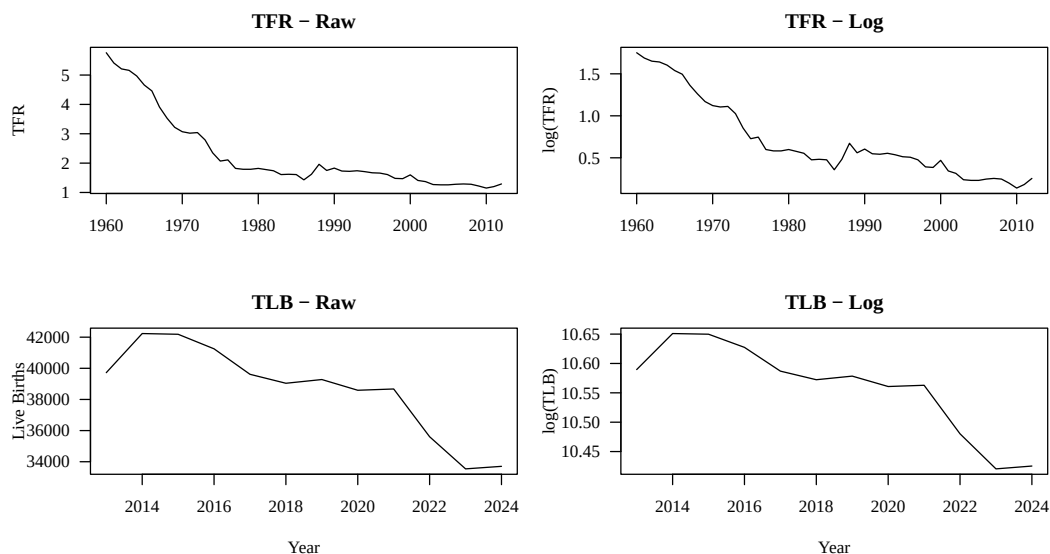


Figure 2: Raw versus log-transformed series for TFR (top row) and TLB (bottom row); training period 1960–2012.

Figure 2 confirms that the log-transformation reduces heteroskedasticity considerably among both series; the large fluctuations earlier on in TLB are quite attenuated in comparison to the stable post-1985 period. However, a downward trend continues among both series, which prompts for first differencing.

4.2 Log-Differenced Series

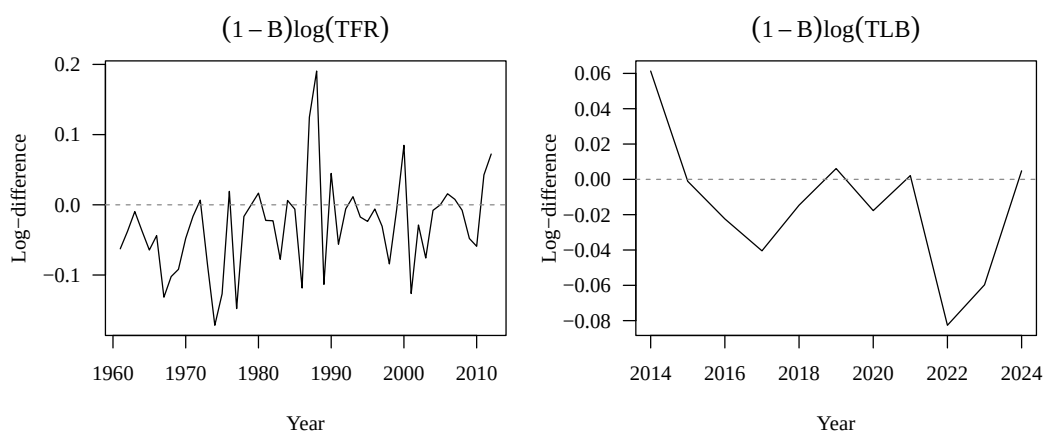


Figure 3: Log-differenced series $(1-B)\log X_t$ for TFR (left) and TLB (right), over training period 1961–2012.

4.3 Stationary Tests

4.3.1 Raw Series

Table 2: Stationarity tests on raw training series (1960–2012).

Series	ADF p-value	KPSS p-value	Conclusion
TFR	0.1200	0.0100	Non-stationary
TLB	0.6541	0.0552	Non-stationary

4.3.2 Log-Differenced series

Table 3: Stationarity tests on log-differenced training series.

Series	ADF p-value	Conclusion
$(1-B)\log(\text{TFR})$	0.0312	Stationary
$(1-B)\log(\text{TLB})$	0.1000	Stationary

4.4 ACF & PACF Analysis

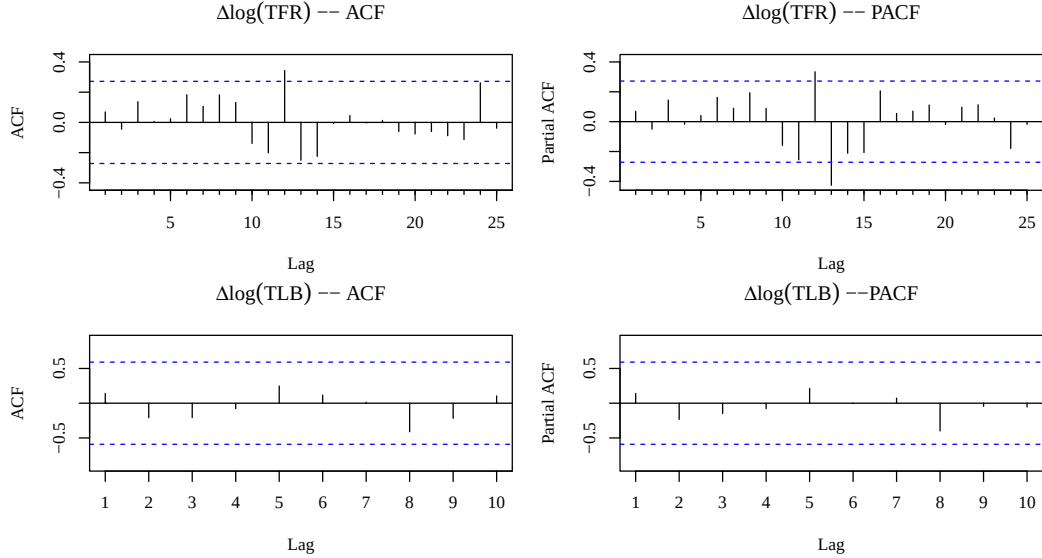


Figure 4: ACF (left column) and PACF (right column) of $(1 - B) \log X_t$ for TFR (top row) and TLB (bottom row), over training period. Lags extend to 25. Blue dashed lines are 95% bounds $\pm 1.96/\sqrt{n}$.

5. Model Identification and Estimation

5.1 TFR Candidate Model

- **T1:** ARIMA(0, 1, 1); pure MA(1) on the log-differenced series;
- **T2:** ARIMA(1, 1, 1); ARMA(1, 1) on the log-differenced series;
- **T3:** ARIMA(1, 1, 0); pure AR(1) on the log-differenced series.

Table 4: Information criteria for TFR candidate models (log scale).

Model	AIC	AICc	BIC
ARIMA(0, 1, 1)	-129.171	-128.671	-123.317
ARIMA(1, 1, 1)	-127.586	-126.735	-119.781
ARIMA(1, 1, 0)	-129.137	-128.637	-123.283

The model with the lowest AICc (bolded in Table 4, for reference) is selected. Its estimated coefficients are printed below:

Table 5: Estimated parameters: ARIMA(0, 1, 1) (log-scale TFR).

Parameter	Estimate	Std. Error	z-stat	p-value
ma1	0.08201	0.15195	0.540	0.5894
drift	-0.02868	0.00988	-2.902	0.0037

5.2 TLB Candidate Model

- **L1:** ARIMA(0, 1, 1);
- **L2:** ARIMA(1, 1, 1);
- **L3:** ARIMA(0, 1, 2) (two MA terms to identify lag-2 structure).

Table 6: Information criteria for TLB candidate models (log scale).

Model	AIC	AICc	BIC
ARIMA(0, 1, 1)	-36.534	-33.106	-35.341
ARIMA(1, 1, 1)	-35.434	-28.767	-33.842
ARIMA(0, 1, 2)	-35.875	-29.208	-34.283

Table 7: Estimated parameters: ARIMA(0, 1, 1) (log-scale TLB).

Parameter	Estimate	Std.Error	z_stat	p-value
ma1	0.40461	0.40262	1.005	0.3149
drift	-0.01157	0.01460	-0.792	0.4281

6 Residual Diagnostics

6.1 TFR Model

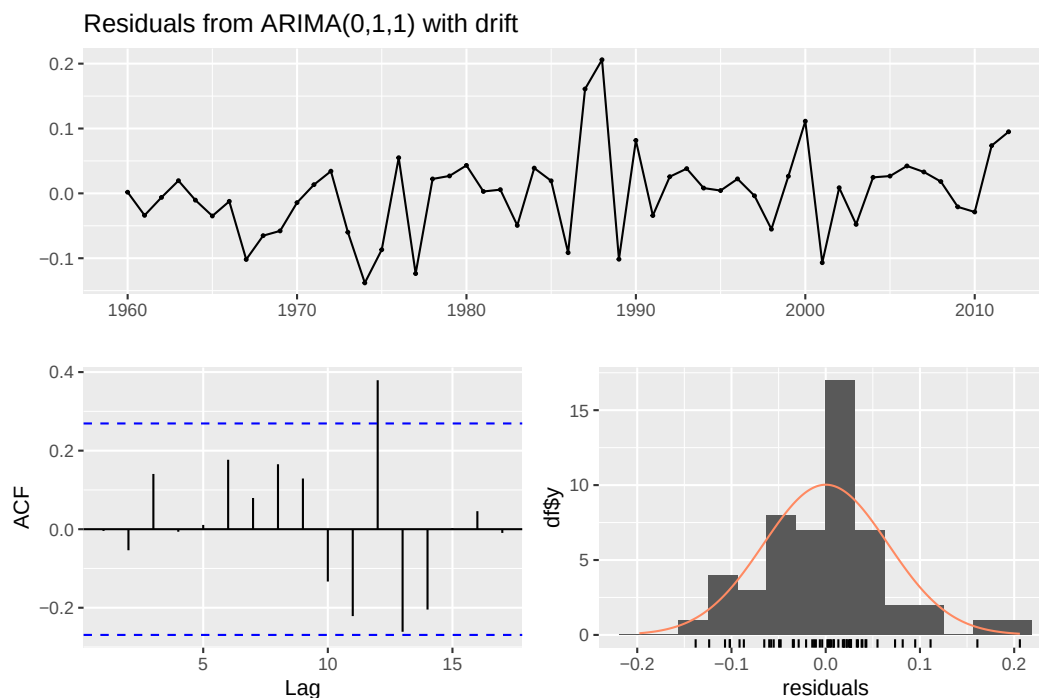


Figure 5: Residual diagnostics for the selected TFR model (ARIMA(0, 1, 1)).

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,1,1) with drift
## Q* = 30.331, df = 19, p-value = 0.04773
##
## Model df: 1.   Total lags used: 20
```

Table 8: Ljung-Box test on TFR model residuals; lag 20, fitdf = $p + q$.

Model	Test lag	fitdf	Q*	df	p-value
ARIMA(0, 1, 1)	20	1	30.331	19	0.0477

Report GitHub Repo

https://github.com/XCalibr3S/TSA_EDA.git

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